

Name: _____ Date: _____

Period: _____

Primary Objective**I will be able to...**

- LO 3.7.A** — Construct a sinusoidal model from a data set by computing amplitude, midline, period, and phase shift, and validate the model against known data values. _____
- LO 3.7.A** — Use a sinusoidal model to predict output values and identify the contextual domain over which the model is meaningful. _____

Vocabulary & Key Concepts

sinusoidal model: A function of the form $f(x) = A \sin(B(x - C)) + D$ (or cosine form) used to represent a _____ real-world quantity.

amplitude: $A = \frac{\max - \min}{2}$; measures how far the function deviates from its _____.

vertical shift / midline: $D = \frac{\max + \min}{2}$; the horizontal line $y = D$ about which the function _____.

period: The _____ input-value interval between consecutive maxima (or minima) in the data. Equals $\frac{2\pi}{B}$.

phase shift: C ; shifts the graph _____ when $C > 0$. Determined by comparing a data maximum to the standard model maximum location.

contextual domain: The set of input values over which the model is _____ given the real-world context.

Hook

Entry Question: The table below shows average monthly high temperatures ($^{\circ}\text{F}$) for a city over one year.

Month x	1	2	3	4	5	6	7	8	9	10	11	12
Temp $T(x)$ ($^{\circ}\text{F}$)	38	42	54	65	74	82	86	84	75	63	49	40

Q1. Which month has the maximum temperature? _____ Which has the minimum? _____

Q2. What kind of function could model this data? _____

LO 3.7.A — Building a Sinusoidal Model: Four Steps

We use the model $T(x) = A \sin(B(x - C)) + D$ where:

- A = amplitude, D = midline (vertical shift),

- $B = \frac{2\pi}{k}$ (k = period), C = phase shift.

Step 1 — Find the Period k

The period equals the input-value distance between consecutive _____ (or consecutive _____).

In the temperature table, consecutive maxima occur at months _____ and _____ (next year), so $k \approx$ _____.

$$B = \frac{2\pi}{k} = \underline{\hspace{2cm}}$$

Step 2 — Compute Amplitude and Midline

$$\text{max} = \underline{\hspace{2cm}} \quad \text{min} = \underline{\hspace{2cm}}$$

$$A = \frac{\text{max} - \text{min}}{2} = \frac{\underline{\hspace{2cm}} - \underline{\hspace{2cm}}}{2} = \underline{\hspace{2cm}}$$

$$D = \frac{\text{max} + \text{min}}{2} = \frac{\underline{\hspace{2cm}} + \underline{\hspace{2cm}}}{2} = \underline{\hspace{2cm}}$$

The midline represents _____ in context.

Step 3 — Determine the Phase Shift C

For $T(x) = A \sin(B(x - C)) + D$, the maximum occurs when $B(x - C) = \frac{\pi}{2}$, so $x = C + \frac{\pi}{2B}$.

The data maximum occurs at $x = \underline{\hspace{2cm}}$. Set equal and solve for C :

$$C + \frac{\pi}{2B} = \underline{\hspace{2cm}} \quad \Rightarrow \quad C \approx \underline{\hspace{2cm}}$$

Tip: If you use a **cosine** model $T(x) = A \cos(B(x - C)) + D$, the maximum occurs when $B(x - C) = 0$, so $C = x_{\text{max}} \underline{\hspace{2cm}}$.

Step 4 — Write and Validate the Model

$$T(x) = \underline{\hspace{4cm}}$$

Validate: substitute a known data point (e.g., $x = 1$, $T = 38$):

$$T(1) \approx \underline{\hspace{2cm}} \quad \text{Residual} = \text{actual} - \text{predicted} = \underline{\hspace{2cm}}$$

Contextual Domain: This model is valid for _____.

Guided Practice

Guided Practice. The table below shows the depth of water (in feet) at a harbor entrance over one tidal cycle (12 hours).

Time t (hr)	0	2	4	6	8	10	12
Depth $d(t)$ (ft)	5	9.5	13	9.5	5	0.5	5

(a) Identify $\text{max} = \underline{\hspace{2cm}}$ at $t = \underline{\hspace{2cm}}$ and $\text{min} = \underline{\hspace{2cm}}$ at $t = \underline{\hspace{2cm}}$.

(b) Compute: $A = \underline{\hspace{2cm}}$ $D = \underline{\hspace{2cm}}$

(c) Estimate the period $k = \underline{\hspace{2cm}}$ hr. Then $B = \frac{2\pi}{k} = \underline{\hspace{2cm}}$.

(d) Identify the phase shift C , using a cosine model with maximum at $t = 4$. $C = \underline{\hspace{2cm}}$.

(e) Write the model: $d(t) = \underline{\hspace{4cm}}$

(f) Use your model to predict the depth at $t = 3$. _____

(g) Is $t = 15$ hr inside the contextual domain for one tidal cycle? Explain. _____